

Immunohematology

Original MCQ Reviewer with Answer Key and Rationales

55 newly written questions covering the same core concepts and section structure as the supplied Chapter 4 study material. Wording and scenarios are original rather than copied from the source.

Sections: Genetics & Immunology • ABO • Rh • Antibody Testing • Compatibility • Transfusion Reactions • Components • Donors • HDFN • Serological Testing • Problem-Solving

4.1 Genetics and Immunology of Blood Groups

1. A blood bank reports a patient as group A after testing the antigens expressed on the red cells. What type of information is being determined?

- A. Genotype
- B. Phenotype
- C. DNA sequence
- D. Chromosome count

2. An antibody reacts 4+ with cells carrying two copies of an antigen but only 2+ with cells carrying one copy. Which phenomenon best explains the difference?

- A. Dosage
- B. Chimerism
- C. Linkage equilibrium
- D. Recessive inheritance

3. Which inheritance pattern best describes the simultaneous expression of A and B antigens in an AB individual?

- A. Complete dominance
- B. Recessive inheritance
- C. Codominance
- D. Sex-linked recessiveness

4. Which genotype is required for expression of the classic Bombay phenotype?

- A. HH
- B. Hh
- C. hh
- D. Either HH or Hh

5. In the Hardy-Weinberg expression $p^2 + 2pq + q^2$, which term represents heterozygotes?

- A. p^2
- B. $2pq$
- C. q^2
- D. $p + q$

4.2 ABO Blood Group System

6. Which finding can distinguish some A2 individuals from A1 individuals?

- A. A2 cells never react with anti-A
- B. A2 individuals may produce anti-A1
- C. A1 individuals routinely produce anti-A2
- D. A1 cells lack H antigen

7. A patient appears group O on forward typing, but the serum agglutinates ordinary group O reagent cells. The patient cells also fail to react with anti-H lectin. Which phenotype is most likely?

- A. A2
- B. Acquired B
- C. Bombay
- D. Weak D

8. Which ABO phenotype generally has the smallest amount of residual H antigen on its red cells?

- A. O
- B. A2
- C. A1
- D. A1B

9. Forward typing gives anti-A 4+ and anti-B 1+, while reverse typing behaves like group A. What should be considered?

- A. Acquired B phenomenon
- B. Bombay phenotype
- C. Rhnull phenotype
- D. Anti-D interference

10. All forward and reverse ABO reactions, including the autocontrol, are positive. Which problem should be investigated first?

- A. Cold autoagglutination
- B. Complete absence of ABO antigens
- C. Bombay phenotype
- D. Loss of all ABO antibodies

4.3 Rh Blood Group System

11. A person lacks D antigen but expresses c and e. How is the person classified for routine Rh status?

- A. Rh positive
- B. Rh negative
- C. Rhnull
- D. Weak D positive

12. Why should weak D testing generally not be interpreted on red cells with a positive DAT?

- A. The cells may already be coated with IgG and produce misleading AHG reactivity
- B. D antigen is destroyed by DAT reagents
- C. Complement removes D antigen
- D. Weak D can only be tested at room temperature

13. A recipient has clinically significant anti-c. Which donor phenotype is acceptable with respect to c?

- A. A unit that expresses c strongly
- B. A unit lacking c antigen
- C. Any D-negative unit
- D. Any C-negative unit

14. Which Rh phenotype is associated with unusually strong D expression because other major Rh antigens are absent?

- A. Rhnull
- B. D⁺-/D⁺-
- C. dce/dce
- D. DCe/dce

15. Which combination is required for classic serologic weak D testing?

- A. Room temperature plus immediate spin
- B. 37 C incubation plus IAT
- C. 4 C incubation plus DAT
- D. Enzyme treatment only

4.4 Testing for Antibodies

16. A Lewis antibody is obscuring a clinically important antibody in a patient sample. Which approach can selectively remove anti-Lea activity?

- A. Neutralization with appropriate saliva
- B. Heating serum to 56 C
- C. Treating cells with DTT only
- D. Adding anti-human globulin before incubation

17. A suspected Donath-Landsteiner antibody binds red cells in the cold and produces hemolysis after warming. What type of behavior is this?

- A. Biphasic hemolysis
- B. Rouleaux formation
- C. Warm panagglutination
- D. Dosage

18. An antibody reacts with essentially all adult screening cells but is weak or negative with cord cells. Which specificity is suggested?

- A. Anti-I
- B. Anti-i
- C. Anti-K
- D. Anti-Fya

19. What does satisfying the 3+3 rule support during antibody identification?

- A. Approximately 95% confidence in the proposed specificity
- B. Proof that the antibody is IgM
- C. Exclusion of all dosage effects
- D. Confirmation of a positive DAT

20. A patient has a documented history of anti-Jka, but today the antibody screen is negative. What is the safest transfusion approach?

- A. Ignore the historical antibody
- B. Select Jka-negative compatible units
- C. Use only group O units regardless of Jka
- D. Repeat testing only after transfusion

4.5 Compatibility Testing

21. During a life-threatening emergency, uncrossmatched O-negative RBCs are released before a patient sample can be obtained. What should happen once a sample becomes available?

- A. No further testing is needed
- B. Compatibility testing should be completed as soon as possible
- C. Only donor plasma should be tested
- D. The emergency units must automatically be discarded

22. All three antibody screening cells react at IAT, but the autocontrol is negative. Which explanation is most plausible?

- A. Warm autoantibody
- B. High-frequency alloantibody or multiple alloantibodies
- C. Rouleaux only
- D. ABO subgroup only

23. What specimens are combined in a minor crossmatch?

- A. Recipient plasma with donor RBCs
- B. Recipient RBCs with donor plasma
- C. Donor RBCs with donor plasma
- D. Recipient RBCs with recipient plasma

24. Six donor units are tested. Five are compatible, one is incompatible, and the patient antibody screen is negative. What should be considered for the incompatible unit?

- A. A positive DAT on the donor cells
- B. A high-frequency patient alloantibody
- C. A universal warm autoantibody
- D. An ABO discrepancy affecting every unit

25. When may an immediate-spin crossmatch be used according to the chapter?

- A. Whenever the patient is Rh positive
- B. When the current screen is negative and there is no antibody history
- C. Only during massive transfusion
- D. Only for group O recipients

4.6 Transfusion Reactions

26. After a suspected transfusion reaction, the technologist gathers the unit, specimens, and records. What is the next priority?

- A. Perform an antibody panel immediately
- B. Check for clerical or identification errors
- C. Measure donor hematocrit
- D. Repeat infectious disease testing on the donor

27. A hemolytic reaction occurs about a week after transfusion. Which mechanism best fits?

- A. Delayed immune response from a previously formed alloantibody
- B. Immediate ABO mismatch only
- C. Simple volume overload
- D. Nonimmune mechanical hemolysis during collection

28. Which antibody type is commonly associated with TRALI in the chapter?

- A. Anti-IgA
- B. Antileukocyte antibody
- C. Anti-A1
- D. Anti-H

29. A patient develops a temperature increase greater than 1 C during transfusion without evidence of hemolysis. Which reaction is most consistent?

- A. Febrile nonhemolytic transfusion reaction
- B. Acute hemolytic reaction
- C. Anaphylactic reaction
- D. Delayed hemolytic reaction

30. A patient develops tachycardia and marked hypertension after receiving a substantial RBC volume, with a negative DAT and little temperature change. Which reaction is most likely?

- A. Volume overload
- B. TRALI
- C. Acute hemolysis
- D. Delayed hemolysis

4.7 Blood Components

31. A patient has hemoglobin of 6 g/dL and the goal is approximately 12 g/dL. Using the chapter rule of thumb, about how many RBC units may be needed?

- A. 2
- B. 3
- C. 6
- D. 10

32. A red cell unit has remained outside controlled 1-6 C storage for more than 30 minutes and is returned to the blood bank. What is the concern?

- A. The unit may no longer meet return-to-inventory requirements
- B. The unit becomes Rh positive
- C. The plasma becomes type AB
- D. The leukocytes are completely removed

33. Which component is emphasized for a patient with chronic granulomatous disease who cannot effectively control severe bacterial infection?

- A. Cryoprecipitate
- B. Granulocyte concentrate
- C. FFP
- D. Washed platelets

34. What storage condition is characteristic of standard platelets?

- A. Frozen below -18 C
- B. 1-6 C without movement
- C. 20-24 C with continuous agitation
- D. 37 C in an incubator

35. Why are cellular blood products irradiated for selected high-risk recipients?

- A. To prevent donor lymphocytes from proliferating
- B. To destroy ABO antigens
- C. To remove all plasma proteins
- D. To increase platelet count

4.8 Donors

36. A healthy donor received a hepatitis B vaccine recently and has no symptoms. According to the chapter, what is the donation status?

- A. Automatically permanently deferred
- B. No deferral solely because of the vaccine if symptom-free
- C. Deferred for 12 months
- D. Eligible only for plasma

37. Which plateletpheresis finding would make a frequent platelet donor unacceptable?

- A. Platelet count $75 \times 10^9/L$
- B. Aspirin taken 7 days ago
- C. Platelet donation 4 days ago
- D. Plasma loss of 800 mL one week ago

38. A highly trained endurance athlete has a resting pulse of 45 beats/min but is otherwise healthy. What does the chapter indicate?

- A. The donor may still be acceptable
- B. The donor is permanently deferred
- C. Only plasma may be collected
- D. The donor must receive atropine first

39. Which blood pressure result would cause rejection under the chapter criteria?

- A. 110/70 mm Hg
- B. 130/80 mm Hg
- C. 170/95 mm Hg
- D. 160/110 mm Hg

40. A whole-blood donor is taking clopidogrel. Which component is specifically affected by the medication?

- A. Platelets
- B. RBCs
- C. Cryoprecipitate
- D. Albumin

4.9 Hemolytic Disease of the Fetus and Newborn

41. Why can an Rh-negative pregnant patient have a positive antibody screen without having formed immune anti-D?

- A. Passive anti-D may be present after Rh immune globulin
- B. Rh-negative people naturally make anti-D
- C. Fetal cells always cause anti-K
- D. Pregnancy destroys D antigen

42. An O-negative mother delivers a B-positive infant in her first pregnancy. The infant is mildly jaundiced and DAT positive. Which maternal antibody can cross the placenta and coat the infant cells?

- A. IgG anti-A,B
- B. IgM anti-H only
- C. IgM anti-M only
- D. Anti-I that remains strictly cold reactive

43. Why is a routine antibody screen generally not performed on cord blood?

- A. Neonates do not produce their own significant antibodies at birth
- B. Cord cells lack all blood group antigens
- C. Maternal antibodies cannot cross the placenta
- D. Cord serum always causes rouleaux

44. An AB-positive infant has HDFN caused by anti-D. Which RBC choice can avoid D antigen while remaining ABO compatible?

- A. AB-negative or O-negative RBCs
- B. AB-positive only
- C. B-positive only
- D. A-positive only

45. Maternal anti-E titers continue to rise and fetal distress begins before delivery is appropriate. Which intervention in the chapter can temporarily reduce maternal antibody levels?

- A. Plasmapheresis
- B. Rh immune globulin for anti-E
- C. Plateletpheresis
- D. Saline replacement

4.10 Serological Testing of Blood Products

46. A pooled NAT screen for donor HIV RNA is reactive. What is the next step described in the chapter?

- A. Test the individual donor samples from that pool
- B. Release all units because NAT is only screening
- C. Perform ABO typing only
- D. Repeat the same pool after one year

47. Which group of viruses is listed in the chapter as targets of donor nucleic acid testing?

- A. HIV, HCV, HBV, Zika, and West Nile virus
- B. CMV and EBV only
- C. HTLV only
- D. Influenza and RSV

48. Which hepatitis B marker is identified as an early marker of infection in the chapter?

- A. HBsAg
- B. Anti-HBs
- C. Anti-HBc IgG only
- D. Anti-HBs plus anti-HBc IgG

49. Which donor test is NOT listed as universally required in the chapter?

- A. Anti-CMV
- B. HBsAg
- C. HIV testing
- D. Anti-HTLV I/II

50. A patient receiving daratumumab has panreactivity with reagent RBCs. What explains this interference?

- A. CD38 is present on reagent red cells as well as myeloma cells
- B. The drug converts group O cells to group A
- C. The drug removes all Rh antigens
- D. The drug creates anti-H

4.11 Immunohematology Problem-Solving

51. Forward typing is group A, and a saliva neutralization study neutralizes anti-A and anti-H but not anti-B. How should the secretor result be interpreted?

- A. Group A secretor pattern
- B. Group B secretor pattern
- C. Group A nonsecretor pattern
- D. Bombay nonsecretor pattern

52. A forward type shows mixed-field reactivity with anti-A, while reverse grouping is weakly inconsistent. The patient has not been recently transfused. What is the best general strategy?

- A. Perform expanded ABO discrepancy testing with anti-A,B, anti-H, additional reagent cells, and an autocontrol
- B. Report group AB immediately
- C. Ignore mixed-field reactions
- D. Perform Rh typing only

53. An unexpected anti-M interferes with reverse grouping. Which principle can help resolve the ABO type?

- A. Use methods that minimize cold-reactive anti-M interference
- B. Incubate only at 37 C with no controls and report immediately
- C. Add M-positive cells to strengthen the discrepancy
- D. Assume the patient is group O

54. A suspected anti-Jka is weak and shows dosage. Which approach would be LEAST useful for confirming the specificity?

- A. AET-treated cells
- B. Homozygous Jka-positive selected cells
- C. Enzyme-treated cells
- D. Longer incubation

55. A cold antibody reacts with nearly all adult cells, including the autocontrol, but its strength varies. Which additional cells are especially useful to investigate anti-I?

- A. Cord blood cells
- B. Only K-positive cells
- C. Only D-negative cells
- D. Enzyme-treated platelets

Answer Key and Rationales

4.1 Genetics and Immunology of Blood Groups

1. **B** Routine serologic blood grouping demonstrates the observable antigen expression of the red cells, which is the phenotype.
2. **A** Dosage describes stronger antibody reactivity with homozygous antigen expression than with heterozygous expression.
3. **C** ABO A and B alleles are codominant, so both antigens are expressed when both alleles are inherited.
4. **C** The Bombay phenotype lacks H substance and is associated with the hh genotype.
5. **B** The heterozygous population is represented by 2pq; p^2 and q^2 represent the two homozygous populations.

4.2 ABO Blood Group System

6. **B** Some A2 individuals can form anti-A1, whereas an A1 individual would not normally make antibody against A1 on their own cells.
7. **C** Bombay cells lack H antigen and the serum contains anti-H, making ordinary group O red cells incompatible.
8. **D** A1B cells convert much of the available H substance into A and B structures, leaving relatively little H.
9. **A** A weak unexpected anti-B reaction in an otherwise group A pattern can occur with acquired B antigen.
10. **A** Panreactivity involving the autocontrol is consistent with a cold autoantibody; warm washing and adsorption may be needed to resolve the type.

4.3 Rh Blood Group System

11. **B** Routine Rh-positive versus Rh-negative classification depends on the presence or absence of D antigen, not on c or e.
12. **A** Weak D testing includes an antiglobulin phase. Preexisting coating of patient cells can cause a positive reaction unrelated to D antigen.
13. **B** Patients with alloanti-c should receive red cells that lack the corresponding c antigen.
14. **B** D⁺/D⁻ cells can show enhanced D expression because precursor availability is not shared with C/c or E/e expression.
15. **B** Weak D detection uses warm incubation followed by the indirect antiglobulin test to demonstrate weakly expressed D.

4.4 Testing for Antibodies

16. **A** Lewis substances in suitable saliva can neutralize Lewis antibodies and help uncover additional clinically significant specificities.
17. **A** The Donath-Landsteiner antibody characteristically binds at cold temperature and causes complement-mediated hemolysis after warming to 37 C.

18. A Adult red cells are rich in I antigen, whereas cord cells predominantly express i antigen, so anti-I reacts poorly with cord cells.

19. A The 3+3 rule uses reactive antigen-positive cells and nonreactive antigen-negative cells to support antibody identification at about 95% confidence.

20. B Kidd antibodies may fall below detectable levels. A known historical anti-Jka must still be honored with antigen-negative blood.

4.5 Compatibility Testing

21. B Emergency release does not eliminate the need for compatibility testing. Testing should proceed when a patient specimen becomes available.

22. B Panreactivity with reagent cells plus a negative autocontrol points away from an autoantibody and toward alloantibody reactivity.

23. B The minor crossmatch tests recipient red cells against donor serum or plasma.

24. A An isolated incompatible donor with a negative screen can result from antibody or complement already coating that donor unit.

25. B The abbreviated immediate-spin approach is appropriate when there is no history of clinically significant antibodies and the current antibody screen is negative.

4.6 Transfusion Reactions

26. B Clerical and identification errors are an important cause of serious transfusion reactions and should be checked early in the investigation.

27. A A delayed hemolytic transfusion reaction can follow an anamnestic rise in antibodies such as Kidd antibodies several days after exposure.

28. B TRALI is associated with antibodies directed against leukocyte antigens, including HLA or neutrophil antigens.

29. A A transfusion-associated fever rise of more than 1°C without another explanation is characteristic of a febrile nonhemolytic reaction.

30. A The hemodynamic pattern and transfused volume support transfusion-associated circulatory overload rather than immune hemolysis.

4.7 Blood Components

31. C One RBC unit typically raises hemoglobin by roughly 1 to 1.5 g/dL, so an increase of about 6 g/dL may require approximately six units.

32. A The chapter applies a 30-minute limit to RBC units kept outside appropriate storage conditions before return to inventory.

33. B Granulocyte concentrates may provide short-term support when phagocytic function is severely impaired.

34. C Standard platelets are stored at room-temperature range with continuous agitation to preserve function.

35. A Irradiation prevents transfusion-associated graft-versus-host disease by disabling the proliferative capacity of donor lymphocytes.

4.8 Donors

- 36. B** The chapter states that a symptom-free donor does not require a deferral simply because of hepatitis B vaccination.
- 37. A** The chapter gives a minimum platelet count above $150 \times 10^9/L$ for frequent platelet donors; $75 \times 10^9/L$ is too low.
- 38. A** Athletes can have physiologic bradycardia and may remain eligible when the low pulse reflects conditioning rather than illness.
- 39. D** The chapter lists a diastolic upper limit of 100 mm Hg, so a diastolic pressure of 110 mm Hg is unacceptable.
- 40. A** Clopidogrel impairs platelet function, so platelet products should not be collected for transfusion from that donor under the chapter guidance.

4.9 Hemolytic Disease of the Fetus and Newborn

- 41. A** Rh immune globulin can produce detectable passive anti-D and is a common reason for a positive screen in an Rh-negative pregnant patient.
- 42. A** The chapter identifies maternal IgG anti-A,B as a possible cause of a positive neonatal DAT in this setting.
- 43. A** Antibodies found in a newborn are maternally derived; the chapter notes that babies do not begin producing their own antibodies until later infancy.
- 44. A** AB-negative or O-negative red cells lack D and are ABO compatible choices for an AB infant in the scenario described.
- 45. A** Plasmapheresis can lower circulating maternal anti-E temporarily while fetal maturity and disease severity are assessed.

4.10 Serological Testing of Blood Products

- 46. A** Reactive NAT pools require resolution by testing individual donor samples to identify the reactive donation.
- 47. A** The chapter lists NAT for HIV, HCV, HBV, Zika, and West Nile virus.
- 48. A** The chapter identifies hepatitis B surface antigen as the first listed marker to appear in HBV infection.
- 49. A** CMV antibody testing is not required for every donor unit, though it may be used for selected high-risk recipients.
- 50. A** Daratumumab targets CD38, which is also expressed on red cells, producing broad serologic reactivity that can mimic a warm autoantibody pattern.

4.11 Immunohematology Problem-Solving

- 51. A** Neutralization of anti-A and anti-H indicates soluble A and H substances in saliva, supporting a group A secretor.
- 52. A** The chapter recommends a structured discrepancy workup rather than assigning an ABO type before the conflicting reactions are explained.
- 53. A** Anti-M is often cold reactive. Techniques that reduce or remove its cold reactivity can clarify the expected reverse-grouping reactions.

54. A The chapter notes that AET treatment denatures Kell antigens but does not specifically enhance or clarify Kidd antibodies, while homozygous cells, enzymes, and longer incubation can help detect anti-Jka.

55. A Cord red cells predominantly express i rather than I antigen, so anti-I should react weakly or not at all and can be distinguished from other cold-reactive patterns.

Source note: This reviewer was created from the concepts and organization of the Immunohematology chapter supplied in this conversation. The questions, choices, and rationales were newly phrased for study use.